Bridging the Gap Between Quantitative Rankings and Qualitative Assessment in University Sustainability: A Framework for Italian Universities

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1 Introduction

Higher Education Institutions (HEIs) are currently navigating a complex landscape where they must reconcile technical efficiency with the mandates of the United Nations' 2030 Agenda and the 17 Sustainable Development Goals (SDGs) (United Nations, 2024). As universities face simultaneous pressures from budget constraints and growing societal expectations, sustainability has transitioned from a peripheral concern to a core institutional priority. Consequently, HEIs are increasingly assessed not only on their operational productivity but on their capacity to act as catalysts for societal change through Environmental, Social, and Governance (ESG) criteria.

Building upon the findings of our Systematic review, which identified a significant "methodological schism" between purely mathematical efficiency models and qualitative sustainability benchmarks, this project aims to bridge this gap. Traditionally, university evaluation has been dominated by non-parametric frontier methods, such as Data Envelopment Analysis (DEA), which focus on quantifiable inputs and outputs like funding, staff, and publications. While robust for benchmarking, these models often reward a "doing more with less" approach that can conflict with long-term sustainability goals (King et al., 2026).

This paper presents an exploratory comparative analysis between existing efficiency and sustainability measures. We utilise institutional productivity data from the Sapientia Observatory (Sapienza Università di Roma, 2026) in conjunction with sustainability metrics extracted from two sustainability global ranking systems: the THE Impact Rankings and UI GreenMetric (Times Higher Education, 2025; Universitas Indonesia, 2025). Our objective is to identify potential cross-drivers - defined here as the reciprocal influences where efficiency indicators affect sustainability performance and vice versa. To identify these, we intentionally employ a straightforward correlation analysis rather than more complex mathematical modelling such as Data Envelopment Analysis (DEA). This methodological choice is deliberate; our focus is not on the quantitative rankings themselves[1], but on bridging the gap between these numbers and the subsequent qualitative assessments.

By identifying these cross-drivers, we categorise universities into four distinct performance clusters based on their efficiency-sustainability profiles: (1) High Efficiency-High Sustainability, (2) High Efficiency-Low Sustainability, (3) Low Efficiency-High Sustainability, and (4) Low Efficiency-Low Sustainability. These groups, along with the cross-drivers identified through our quantitative analysis, then direct the qualitative stage of our research. In this phase, we analyze sustainability reporting and disclosure levels to investigate the institutional practices associated with the observed performance patterns. Using Italian universities as a case study, we demonstrate how quantitative results lead directly to meaningful qualitative assessments through two proposed methods: driver identification and sustainability reporting analysis.

[1] We believe the methodology in this paper can be conducted on any other (and more advanced) domain of analysis but in an effort to control the experiment, and not let the bridging to qualitative assessment be dependent on the the method of quantitative analysis, we chose the simplest of quantitative experiments.

2 Literature review

2.1 Systematic review methodology

The study is based on the earlier conducted systematic literature review on 435 articles on the interaction between the aspects of technical efficiency and sustainability by HEIs (King et al., 2026). That work revealed that current research relies almost entirely on measurable inputs and outputs, often ignoring the qualitative factors that drive institutional change. Thus, an evident gap appears where operational efficiency is measured, while the sustainable outcome of the process is overlooked.

The systematic review concluded that current university evaluation methods are split between mathematical efficiency and sustainability metrics, with almost no connection between them. Because these two areas are not aligned, existing rankings often provide an incomplete or misleading picture of a university's actual performance. The findings show that without a more integrated model, sustainability is frequently treated as a symbolic gesture rather than a core part of institutional operations.

2.2 The UN Sustainable Development Goals (SDGs)

The primary framework for assessing global sustainability within HEIs in this project is the United Nations 2030 Agenda, which comprises 17 Sustainable Development Goals (SDGs) (United Nations, 2024). A detailed overview of each individual goal and the specific metrics we have extracted for our analysis is provided in the following section. Literature suggests that HEIs must move beyond symbolic adoption and incorporate these SDGs directly into core performance metrics to act as primary catalysts for societal change (Lozano et al., 2015; Malešević Perović and Mihaljević Kosor, 2020; Fiorani and Di Gerio, 2022).

2.3 Sustainability rankings

The UI GreenMetric and THE Impact Rankings serve as the primary benchmarks for this analysis, though they diverge significantly in their evaluative scope (Universitas Indonesia, 2025; Times Higher Education, 2025). Semantically, UI GreenMetric focuses on the **environmental and operational** dimensions of a "green campus," prioritising physical infrastructure and resource management (Lauder et al., 2015). In contrast, the THE Impact Rankings adopt a **multidimensional societal** lens, measuring institutional contribution across the 17 SDGs (Bautista-Puig et al., 2022). While the former benchmarks ecological efficiency at the campus level, the latter evaluates broader impact profiles through a combination of social, economic, and environmental targets.

2.4 The role of sustainability reporting

Sustainability reporting, primarily through the "Bilancio di Sostenibilità" (Fiorani and Di Gerio, 2022), serves as a fundamental mechanism for institutional accountability and transparency. These reports are not merely data disclosure tools; they act as instruments to legitimise and reinforce organisational change within the university structure.

In the Italian context, reporting practices often adhere to international standards such as the Global Reporting Initiative (GRI) or local frameworks like the RUS-GBS standard (Ezza et al., 2024). Data from the GBS Observatory indicates that while a majority of Italian universities maintain official strategic plans, a significant gap remains in the adoption of dedicated Sustainability Plans (Fiorani and Di Gerio, 2022). This inconsistency, identified in our systematic synthesis, underscores the challenge of translating high-level sustainability commitments into coordinated, measurable institutional action.

3 Data collection

This section details the integration of institutional performance metrics with sustainability benchmarks. To ensure a robust comparative analysis, the dataset was standardised using 2023 as the baseline year. For each variable, we utilized a baseline value alongside 1-year, 2-year, and 3-year differences, as well as an acceleration factor.

3.1 Efficiency data (Sapientia Observatory)

The efficiency metrics were sourced from the Sapientia Observatory database. We utilised the primary data sheet, which tracks the resource inputs and academic outputs of Italian Higher Education Institutions (HEIs).

Although the initial Sapientia database included 60 HEIs, this study is restricted to the intersection of universities present in all three sources (Sapientia, UI GreenMetric, and THE Impact Rankings). This filtering resulted in a final sample of **29 universities**:

- Università degli Studi di Bari Aldo Moro
- Università degli Studi della Basilicata
- Università degli Studi di Bergamo
- Alma Mater Studiorum – Università di Bologna
- Università degli Studi di Cagliari
- Università di Camerino
- Università degli Studi "G. d'Annunzio" Chieti – Pescara
- Università degli Studi di Ferrara
- Università degli Studi di Firenze
- Università degli Studi di Foggia
- Università degli Studi di Genova
- Università degli Studi dell'Insubria
- Università degli Studi dell'Aquila
- Università degli Studi di Macerata
- Università Politecnica delle Marche
- Università degli Studi di Milano
- Università degli Studi di Milano-Bicocca
- Politecnico di Milano
- Università degli Studi di Modena e Reggio Emilia
- Università degli Studi di Padova
- Università degli Studi di Parma
- Università degli Studi del Piemonte Orientale "Amedeo Avogadro"
- Università degli Studi Roma Tre
- Università degli Studi di Salerno
- Università degli Studi di Torino
- Politecnico di Torino
- Università degli Studi della Tuscia
- Università degli Studi di Urbino Carlo Bo
- Università Iuav di Venezia

3.2 Sustainability data

Sustainability performance was taken from two distinct rankings, capturing both operational "green" metrics and broader societal impact.

3.2.1 THE Impact Rankings (SDGs)

The Times Higher Education (THE) Impact Rankings utilize the United Nations' Sustainable Development Goals (SDGs) as a structural framework (Times Higher Education, 2025). While the SDGs provide the underlying goals, THE employs a specific measuring process for each, translating the 17 goals into a set of distinct institutional indicators and weighted metrics to quantify performance. Table 1 outlines how specific goals are quantified within this framework.

SDG	Goal Definition	Quantitative Indicators	Measuring Units / Logic	UN Resource
SDG 1	No Poverty	Low-income student proportion; financial aid spending.	% of students; Local currency per student.	https://sdgs.un.org/goals/goal1
SDG 2	Zero Hunger	Campus food waste; student food insecurity programs.	Tons of waste; Binomial (Yes/No) + evidence.	https://sdgs.un.org/goals/goal2
SDG 3	Good Health	Research on key diseases; health outreach programs.	Bibliometric score; No. of outreach events.	https://sdgs.un.org/goals/goal3
SDG 4	Quality Education	Proportion of first-generation students; lifelong learning.	% of total students; Outreach attendance.	https://sdgs.un.org/goals/goal4
SDG 5	Gender Equality	Women in senior academic positions; female student ratio.	% of senior staff; % of female graduates.	https://sdgs.un.org/goals/goal5
SDG 6	Clean Water	Water consumption per person; water recycling practices.	m^3 per person; % of water reused.	https://sdgs.un.org/goals/goal6
SDG 7	Clean Energy	Energy use per building area; renewable energy share.	kWh/m^2; % of renewable energy.	https://sdgs.un.org/goals/goal7
SDG 8	Decent Work	Proportion of staff on secure contracts; gender pay gap.	% of workforce; % difference in salary.	https://sdgs.un.org/goals/goal8
SDG 9	Innovation	Patents citing university research; income from industry.	No. of patents; Currency per academic staff.	https://sdgs.un.org/goals/goal9
SDG 10	Reduced Inequalities	Proportion of international students from developing nations.	% of total international intake.	https://sdgs.un.org/goals/goal10
SDG 11	Sustainable Cities	Sustainable commuting; affordable housing for staff/students.	% of non-car commutes; Housing capacity.	https://sdgs.un.org/goals/goal11
SDG 12	Consumption	Proportion of waste sent to landfill; plastic reduction.	% recycled; Binomial policy verification.	https://sdgs.un.org/goals/goal12
SDG 13	Climate Action	Carbon footprint per floor area (Scope 1 and 2).	Tonnes of CO_2e/m^2.	https://sdgs.un.org/goals/goal13
SDG 14	Life Below Water	Water discharge quality; support for aquatic ecosystems.	Binomial policy + Research impact.	https://sdgs.un.org/goals/goal14
SDG 15	Life on Land	Land used for sustainable practices; biodiversity research.	% of campus area; Bibliometric score.	https://sdgs.un.org/goals/goal15
SDG 16	Peace & Justice	Institutional transparency; student union participation.	Binomial governance check; % turnout.	https://sdgs.un.org/goals/goal16
SDG 17	Partnerships	Research with low-income countries; SDG report frequency.	% of co-authored papers; Annual vs. Periodic.	https://sdgs.un.org/goals/goal17

Table 1: Detailed Measurement Indicators and UN Framework Links for THE Impact Rankings. Adapted from Times Higher Education (2025).

3.2.2 UI GreenMetric

In contrast to the THE Impact Rankings the UI GreenMetric employs an independent methodology specifically designed to measure environmental commitment (Universitas Indonesia, 2025). The measurement focus for each is detailed in Table 2.

3.3 Reporting data

To complement the quantitative sustainability and efficiency datasets, an automated web-scraping framework was developed to collect publicly available sustainability-related information from Italian university

Cat.	Weight	Quantitative Indicators	Measuring Units / Logic	Max Pts
SI	15%	Campus Setting	Ratio of open space to total area; % of forest/vegetation.	1,500
EC	21%	Energy Efficiency	Electricity usage per total campus population ($kWh/person$); Carbon footprint ($metric\ tonnes\ CO_2e$).	2,100
WS	18%	Waste Treatment	% of organic/inorganic waste treated; Toxic waste disposal; Sewage treatment.	1,800
WR	10%	Water Usage	% of water recycled; Use of water-efficient fixtures (taps/toilets).	1,000
TR	18%	Campus Transit	No. of university-owned vehicles; % of parking area; Pedestrian path density.	1,800
ED	18%	Research & Education	Ratio of sustainability courses to total courses; Research funding for sustainability.	1,800
Total	**100%**	—	**Cumulative Score**	**10,000**

Table 2: Quantitative Scoring Framework for UI GreenMetric. Data derived from `https://greenmetric.ui.ac.id/`

websites. The purpose of the scraper was not to construct a formal ranking system, but rather to provide an exploratory evidence discovery tool capable of identifying sustainability reporting intensity, thematic emphasis, and institutional transparency patterns.

The scraper was implemented in Python using a recursive crawling architecture. Each university was associated with a primary institutional URL alongside manually identified sustainability-related root pages. Manual root assignment was necessary because sustainability content is frequently distributed across heterogeneous university subdomains, including sustainability offices, governance sections, strategic planning pages, and sustainability reports.

The crawler recursively explored sustainability-related pages within each institutional domain using a breadth-first search strategy with depth limitations to prevent uncontrolled traversal. URL exploration was restricted through a set of sustainability-oriented path filters, including terms such as *sustainability*, *SDG*, *governance*, *climate*, *bilancio*, and *environment*. Only URLs matching these sustainability-relevant patterns were retained for further processing.

For each identified webpage, HTML content was extracted and cleaned using BeautifulSoup. In parallel, all linked PDF documents were identified and processed separately using the PyMuPDF library. PDF text extraction enabled the inclusion of sustainability reports, governance documents, climate action plans, strategic plans, and related institutional publications that were not directly accessible through webpage text alone.

The extracted corpus was then subjected to thematic processing. Word frequency analysis was performed after removing common stopwords and non-informative institutional terminology. In addition, a lightweight SDG detection framework was implemented through predefined keyword dictionaries corresponding to selected Sustainable Development Goals, including climate action, governance, energy, inclusion, mobility, and education. This enabled the generation of approximate thematic intensity indicators for each institution.

Several summary indicators were subsequently generated, including:

- Number of sustainability-related pages discovered;
- Number of sustainability-related PDF documents;
- Total extracted word count;
- Most frequent sustainability-related terms;
- SDG thematic hit counts;
- Estimated evidence intensity level.

The evidence intensity classification (*none*, *minimal*, *moderate*, *strong*, and *extensive*) was constructed heuristically using a weighted combination of discovered pages, PDF documents, and extracted textual volume. Importantly, this classification should not be interpreted as a direct measure of sustainability performance or institutional quality. Instead, it represents the relative visibility and discoverability of sustainability-related evidence within publicly accessible institutional communication channels.

The scraper also incorporated parallel processing through multithreaded execution in order to improve scalability across universities. Results were exported into both CSV and JSON formats to support downstream quantitative and qualitative analyses.

Despite its usefulness as an exploratory diagnostic framework, several limitations must be acknowledged. First, institutional websites vary substantially in structure, accessibility, and document organisation. Consequently, lower evidence levels may reflect either genuinely limited sustainability reporting or limitations in automated discoverability. Second, the recursive crawl was intentionally constrained through page limits and keyword filtering to ensure computational feasibility, which may have excluded relevant sustainability content hosted outside predefined pathways. Third, PDF extraction quality depends on document formatting and text accessibility, particularly for scanned or image-based files. Finally, the SDG detection framework relies on keyword matching rather than semantic interpretation, meaning that thematic counts should be interpreted as approximate indicators rather than precise measurements.

For these reasons, the scraper should be understood primarily as a subsidiary analytical instrument designed to support institutional sustainability profiling and thematic exploration, rather than as a standalone evaluative methodology.

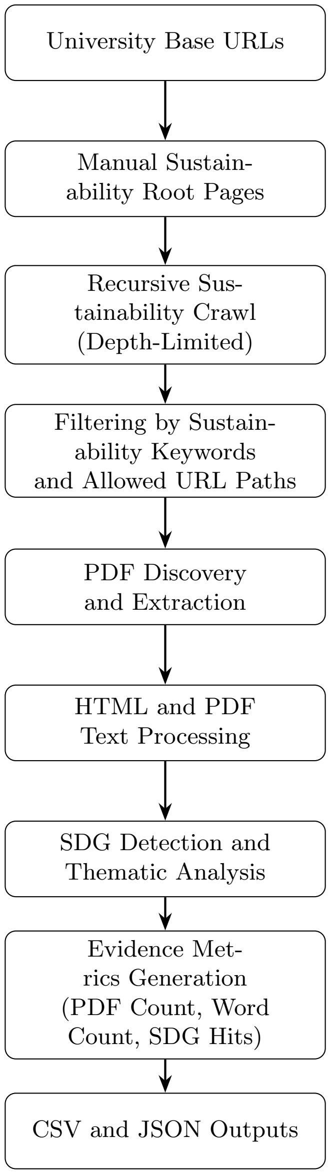

Figure 1: Automated sustainability evidence collection pipeline

4 Exploratory quantitative analysis of efficiency and sustainability metrics

This section presents the results of the Sapientia Observatory's quantitative assessment. Rather than providing definitive causal links, this analysis offers a structured framework to explore potential drivers that influence how institutions coordinate productivity with the integrated sustainability benchmarks previously defined. By examining the intersection of internal efficiency and the multidimensional metrics from both the THE Impact (SDG-based) and UI GreenMetric (operational-based) frameworks, we aim to identify the patterns that characterize the Italian higher education landscape.

4.1 Data Integration and Sustainability Scoring

A key challenge in the data pipeline was the prevalence of missing values (NaNs) found exclusively within the results from the THE Impact dataset. These NaNs exist because universities are not ranked homogeneously across all SDGs in the THE framework (or at least not all of the SDGs are displayed on the website). To ensure the reliability of the results and avoid spurious ± 1 correlations caused by insufficient data points, a filter was applied: only SDGs and UI metrics with a minimum of five entries were included in the analysis. Furthermore, to evaluate the significance of the potential drivers, a *stability score* was calculated. For each efficiency metric, we calculated the mean of its correlations across all other sustainability dimensions. The final stability score was then defined as the ratio of this mean to its standard deviation (μ/σ). This metric allows us to rank the drivers themselves, prioritizing those that demonstrate both high and consistent performance across the sustainability framework.

To map out the immediate interactions within the dataset, an analysis of the top individual correlation pairs was conducted. Given that the variables are systematically structured so that higher values indicate favorable outcomes (e.g., optimized resource management or higher efficiency outputs), positive correlations highlight the synergy.

TOP 50 POSITIVE CORRELATIONS

	Sustainability_Metric	Efficiency_Metric	Correlation
2765	The_Sustainable cities and communities_lower	ECONOMIC AND FINANCIAL_OPERATING COSTS: DEPREC...	0.989502
2761	The_Sustainable cities and communities_lower	ECONOMIC AND FINANCIAL_OPERATING REVENUES: OTH...	0.976858
2875	The_Responsible consumption and production_lower	RESEARCH_ARTICLES PUBLISHED IN NATURE AND SCIE...	0.971894
2784	The_Sustainable cities and communities_lower	RESEARCH_HIGHLY CITED RESEARCHERS _2023	0.967032
2770	The_Sustainable cities and communities_lower	ECONOMIC AND FINANCIAL_TEACHING REVENUES: MAST...	0.962182
2846	The_Responsible consumption and production_lower	ECONOMIC AND FINANCIAL_NET EQUITY_2023	0.954638
2764	The_Sustainable cities and communities_lower	ECONOMIC AND FINANCIAL_OPERATING COSTS: CURREN...	0.951009
2742	The_Sustainable cities and communities_lower	POST LAUREAM EDUCATION_SECOND-LEVEL MASTER'S D...	0.950841
2717	The_Sustainable cities and communities_lower	STAFF_UNIVERSITY STAFF_2023	0.945879
2766	The_Sustainable cities and communities_lower	ECONOMIC AND FINANCIAL_OPERATING COSTS: OTHER_...	0.945410
2762	The_Sustainable cities and communities_lower	ECONOMIC AND FINANCIAL_OPERATING COSTS_2023	0.944148
2732	The_Sustainable cities and communities_lower	STUDENT LIFECYCLE_CREDITS OBTAINED ABROAD_2023	0.941592
2716	The_Sustainable cities and communities_lower	STAFF_NON-ACADEMIC STAFF_2023	0.937258
2703	The_Sustainable cities and communities_lower	STAFF_FULL PROFESSORS_2023	0.936349
2743	The_Sustainable cities and communities_lower	ECONOMIC AND FINANCIAL_STUDENT CONTRIBUTION RE...	0.925385
2753	The_Sustainable cities and communities_lower	ECONOMIC AND FINANCIAL_CURRENT EXPENDITURES_2023	0.921805
2780	The_Sustainable cities and communities_lower	ECONOMIC AND FINANCIAL_PERFORMANCE-BASED FUNDI...	0.920642
2782	The_Sustainable cities and communities_lower	ECONOMIC AND FINANCIAL_EXTRAORDINARY FUNDING P...	0.920641
360	UI Green Metrics_TR	Identifiers_GLOBAL	0.917790

Figure 2: Top 50 Positive Correlations

The strongest positive relationships are heavily dominated by SDG 11 (Sustainable Cities and Communities) and SDG 12 (Responsible Consumption and Production). Specifically, these sustainability areas demonstrate near-perfect alignment with core economic metrics, including institutional operating revenues, operating costs, teaching revenues from master's programs, and the presence of highly cited researchers. This could indicate that financially robust academic environments with prominent research profiles naturally create a supportive baseline for sustainable campus integration and responsible consumption.

Conversely, strong negative correlations expose critical institutional trade-offs, effectively acting as "bad" drivers or operational barriers within the current framework. When interpreting these negative pairs, it is important to note that the dataset incorporates metrics designated with lower, mean, or upper bounds to accurately reflect the top, middle, and bottom values of reported SDG data ranges. To avoid redundancy, these bounds are analyzed as singular overarching trends.

TOP 50 NEGATIVE CORRELATIONS

	Sustainability_Metric	Efficiency_Metric	Correlation
1163	The_Qualitu Education_lower	ECONOMIC AND FINANCIAL_STANDARD COST PER STUDE...	-0.965976
1343	The_Qualitu Education_mean	ECONOMIC AND FINANCIAL_STANDARD COST PER STUDE...	-0.963206
1253	The_Qualitu Education_upper	ECONOMIC AND FINANCIAL_STANDARD COST PER STUDE...	-0.953239
3059	The_Climate Action_upper	RESEARCH_ARTICLES PUBLISHED IN NATURE AND SCIE...	-0.931169
3149	The_Climate Action_mean	RESEARCH_ARTICLES PUBLISHED IN NATURE AND SCIE...	-0.920093
2969	The_Climate Action_lower	RESEARCH_ARTICLES PUBLISHED IN NATURE AND SCIE...	-0.878808
2789	The_Sustainable cities and communities_lower	RESEARCH_ARTICLES PUBLISHED IN NATURE AND SCIE...	-0.803267
2638	The_Reducing inequalities_mean	STUDENT LIFECYCLE_PERCENTAGE OF GRADUATES SATI...	-0.793332
2548	The_Reducing inequalities_upper	STUDENT LIFECYCLE_PERCENTAGE OF GRADUATES SATI...	-0.793020
3413	The_Peace, justice and strong institutions_mean	ECONOMIC AND FINANCIAL_STANDARD COST PER STUDE...	-0.781859
3239	The_Peace, justice and strong institutions_lower	RESEARCH_ARTICLES PUBLISHED IN NATURE AND SCIE...	-0.777840
2726	The_Sustainable cities and communities_lower	STUDENT LIFECYCLE_PERCENTAGE OF GRADUATES WITH...	-0.773737
3323	The_Peace, justice and strong institutions_upper	ECONOMIC AND FINANCIAL_STANDARD COST PER STUDE...	-0.772964
2458	The_Reducing inequalities_lower	STUDENT LIFECYCLE_PERCENTAGE OF GRADUATES SATI...	-0.771064
2371	The_Industry, Innovation, and Infrastricture_mean	STUDENT LIFECYCLE_STUDENT-STAFF RATIO_2023	-0.765507
3236	The_Peace, justice and strong institutions_lower	RESEARCH_ARTICLES PUBLISHED IN NATURE AND SCIE...	-0.764955
2281	The_Industry, Innovation, and Infrastricture_u...	STUDENT LIFECYCLE_STUDENT-STAFF RATIO_2023	-0.760118

Figure 3: Top 50 Negative Correlations

It is shown that SDG 4 (Quality Education) across its entire data range displays a severe negative correlation with the standard cost per student. In additon, SDG 13 (Climate Action) and SDG 11 show sharp negative correlations with the absolute volume of articles published in Nature and Science. This leads us to think that there is a clear divide in how universities operate. When institutions focus too much on achieving elite research rankings and spending large amounts of money per student, they create a major conflict. This intense focus ultimately gets in the way of broader goals like fighting climate change and making education fair for everyone.

4.2 Cross-Correlation Analysis of Potential Drivers

The analysis utilised cross-correlation between sustainability metrics and efficiency-related variables to identify factors that may catalyse or hinder performance. Due to the sample size ($n = 29$ universities), these findings are presented as a proposed framework for further study rather than absolute statistical proof.

4.2.1 Analysis of Positive Drivers

The ten highest-ranked drivers all exhibit positive directional relationships with sustainability performance. As shown in Figure 4, the most significant potential driver is the prevalence of Postgraduate First-Level Master's Degrees, followed closely by Economic-Financial Equalisation Interventions.

The fact that all top drivers show positive coefficients raises an important question for future research: whether the framing of high-performance metrics (where "high" always denotes a positive outcome, such as low waste or high efficiency) implies an inherent correlation between sustainability and technical efficiency in Higher Education Institutions (HEIs).

4.2.2 Negative Drivers and Observations

The analysis also identifies variables that show a negative relationship with sustainability scores. Notably, the number of Articles published in Nature and Science and the average age of staff and PhD graduates emerged as top negative factors. In terms of personnel, results show that higher average ages in the academic body are associated with lower sustainability scores. This suggests that traditional mindsets or older physical infrastructure may create practical barriers to a fast transition toward sustainable practices.

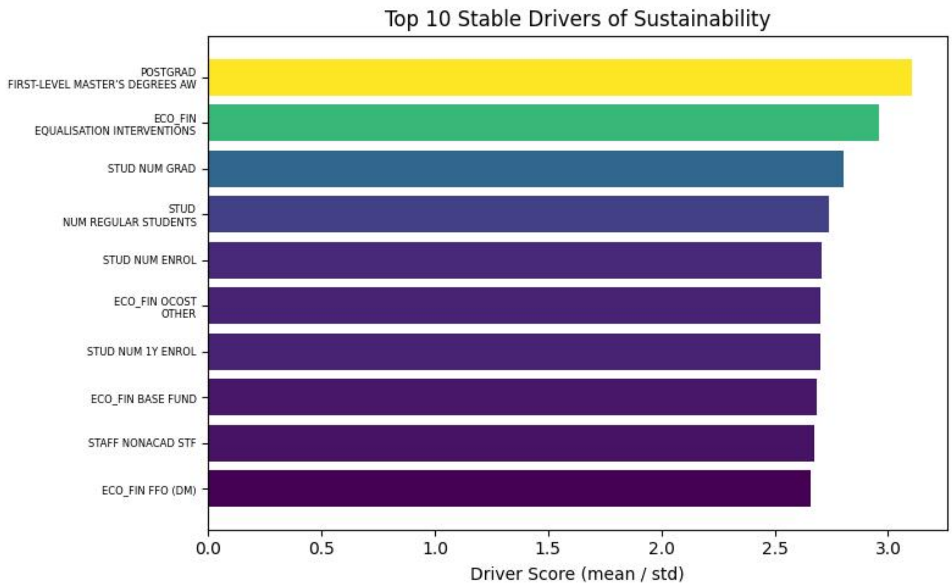

Figure 4: Top 10 Potential Drivers of Sustainability Performance (Stability Analysis).

4.3 The Efficiency vs. Sustainability Matrix

The merged dataset allows for the classification of universities into four quadrants based on their ranked and normalized Efficiency Score (x-axis) and Sustainability Score (y-axis). This visualization summarizes the quantitative output in a way that facilitates qualitative comparison.

- **High Efficiency / High Sustainability (Top-Right):** The "Green" quadrant includes institutions like **Torino Politecnico, Milano Bicocca**, and **Genova**, which currently lead in both operational and ESG metrics.
- **High Efficiency / Low Sustainability (Bottom-Right):** The "Red" cluster (e.g., **Bologna, Padova, Milano Politecnico**) identifies institutions with high traditional productivity that has yet to reflect in top-tier sustainability rankings.
- **Low Efficiency / High Sustainability (Top-Left):** The "Blue" quadrant (e.g., **Urbino, Basilicata**) highlights institutions prioritizing sustainability reporting despite lower technical efficiency.
- **Low Efficiency / Low Sustainability (Bottom-Left):** The "Grey" quadrant (e.g., **Foggia, Insubria**) points to institutions facing structural challenges in both domains.

This matrix serves as a diagnostic tool, enabling university managers to identify whether their institutional gaps are primarily operational or related to sustainability alignment

5 Qualitative contextualisation of quantitative findings

The quantitative analysis identified four institutional clusters based on the relationship between technical efficiency and sustainability performance. The qualitative stage of the research was developed to contextualise these results through the analysis of sustainability reporting, institutional communication, and thematic evidence collected through the web scraping framework described previously, and then from the Word Wall. A *word wall* (or word cloud) is a visual text-analysis representation in which the size of each word is proportional to its frequency within a given corpus of documents. In this study, word walls were generated from the extracted text, larger words therefore indicate terms that appeared more frequently within the scraped sustainability-related communication of a given institutional cluster.

The purpose of the word walls is not to provide semantic interpretation or causal evidence, but rather to offer an exploratory visual summary of the dominant themes and narratives emphasised within institutional reporting. As a result, differences in word prominence reflect variations in communication focus and reporting visibility across university groups.

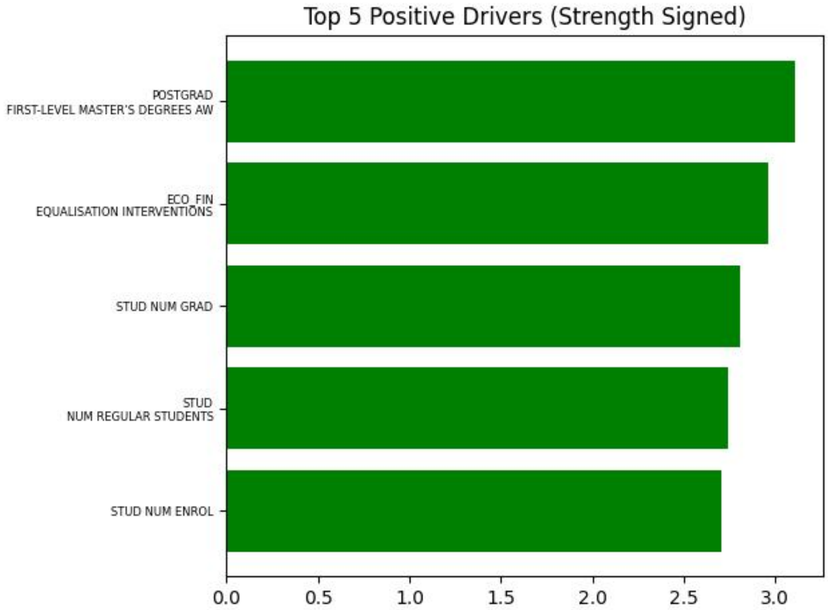

Figure 5: Top 5 factors with the strongest positive relationship to sustainability scores.

Rather than functioning as a causal model, the qualitative analysis acts as an exploratory layer used to interpret how universities communicate sustainability priorities and how these narratives relate to the top identified drivers.

5.1 Cluster comparison through identified drivers

The comparison across quadrants reveals that, for almost all of the top five positive drivers, the High Efficiency–Low Sustainability cluster recorded higher values than the High Efficiency–High Sustainability group (and higher than all the others). This suggests that the universities with the strongest operational productivity are not necessarily those achieving the highest sustainability outcomes. So, institutions in the High Efficiency–Low Sustainability cluster appear to maximise traditional performance indicators, but this operational strength does not automatically translate into a sustainability assessment.

One possible explanation is that highly productive universities may allocate proportionally greater institutional attention and resources toward research competitiveness, international positioning, and conventional performance metrics rather than toward sustainability governance and progressive measures. In this sense, sustainability may remain secondary to traditional measures of academic prestige despite strong institutional capacity.

At the same time, for most of the positive drivers (80%), the Low Efficiency–High Sustainability cluster still performed better than the Low Efficiency–Low Sustainability group. This suggests that even among universities with lower operational efficiency, organisational coordination and sustainability-oriented governance may still contribute positively to sustainability outcomes. The only notable exception was in Equalisation Interventions, where the relationship appeared less consistent across clusters.

A possible interpretation is that sustainability performance is not determined exclusively by institutional scale or productivity, but also by strategic prioritisation and governance orientation. Universities in the Low Efficiency–High Sustainability cluster may compensate for lower technical efficiency through stronger territorial engagement, or more focused institutional commitment to SDG-related initiatives. This reinforces the idea that sustainability integration depends not only on available resources, but also on how institutions choose to organise and communicate their strategic objectives.

The analysis of the negative drivers produced more ambiguous results compared with the positive drivers. Visually, four out of the five strongest negative factors did not display clear or consistent separation across the institutional quadrants. This lack of distinct clustering suggests that many of the negative correlations identified in the quantitative analysis may not represent stable institutional patterns, but rather noisy relationships influenced by limitations in the web scraping and the heterogeneity of Italian HEIs.

In practical terms, this indicates that not all statistically negative relationships necessarily translate

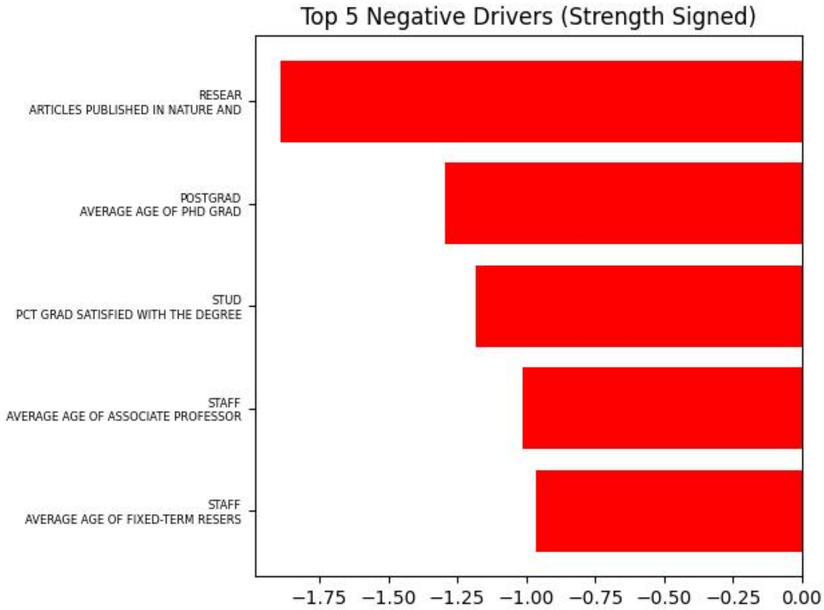

Figure 6: Top 5 factors with a negative relationship to sustainability scores.

into meaningful organisational differences between universities. These cases are equally important for interpreting the framework because they raise questions regarding the stability and interpretability of the identified drivers.

The inconsistency may highlight a limitation of the framework itself when used as a diagnostic instrument. Because the methodology relies on ranked sustainability thresholds and quadrant classification, some institutional characteristics may not distribute cleanly across the groups even if they still contribute indirectly to sustainability outcomes. Consequently, not every identified driver should necessarily be interpreted as equally meaningful from a qualitative perspective.

The most notable exception was the number of *Articles published in Nature*, which showed a more coherent pattern across the quadrants. Universities in the High Efficiency–Low Sustainability cluster consistently displayed a stronger concentration compared with the High Efficiency–High Sustainability group. This finding supports the possibility of a partial trade-off between highly competitive research-oriented institutional models and broader sustainability integration.

But, this does not imply that research excellence itself negatively affects sustainability. Rather, the result may indicate that universities heavily focused on bibliometric prestige and international research competitiveness tend to prioritise traditional academic performance metrics over sustainability governance. In this sense, the relationship may only reflect differences in institutional priorities and resource allocation.

It is also important to note that this exploratory comparison focused only on the five strongest negative drivers for interpretive clarity. However, the proposed framework is not limited to these variables. The same methodological approach can be extended to any identified driver by selecting an appropriate sustainability score threshold.

The graphical comparison of the identified drivers reveals a recurring pattern across many of both the positive and negative variables. In the majority of cases, the High Efficiency–Low Sustainability quadrant recorded the strongest values for efficiency-related drivers, followed by the High Efficiency–High Sustainability group.

One interpretation of this trend is the existence of a partial efficiency–sustainability trade-off. In this sense, extremely strong performance in conventional efficiency indicators may coincide with weaker sustainability alignment.

At the same time, an observation emerges when comparing the two lower-efficiency quadrants. Across most positive drivers, the Low Efficiency–High Sustainability institutions still outperform the Low Efficiency–Low Sustainability group. This suggests that even where operational productivity is comparatively weaker, sustainability-oriented governance and institutional prioritisation may still produce measurable differences in sustainability outcomes.

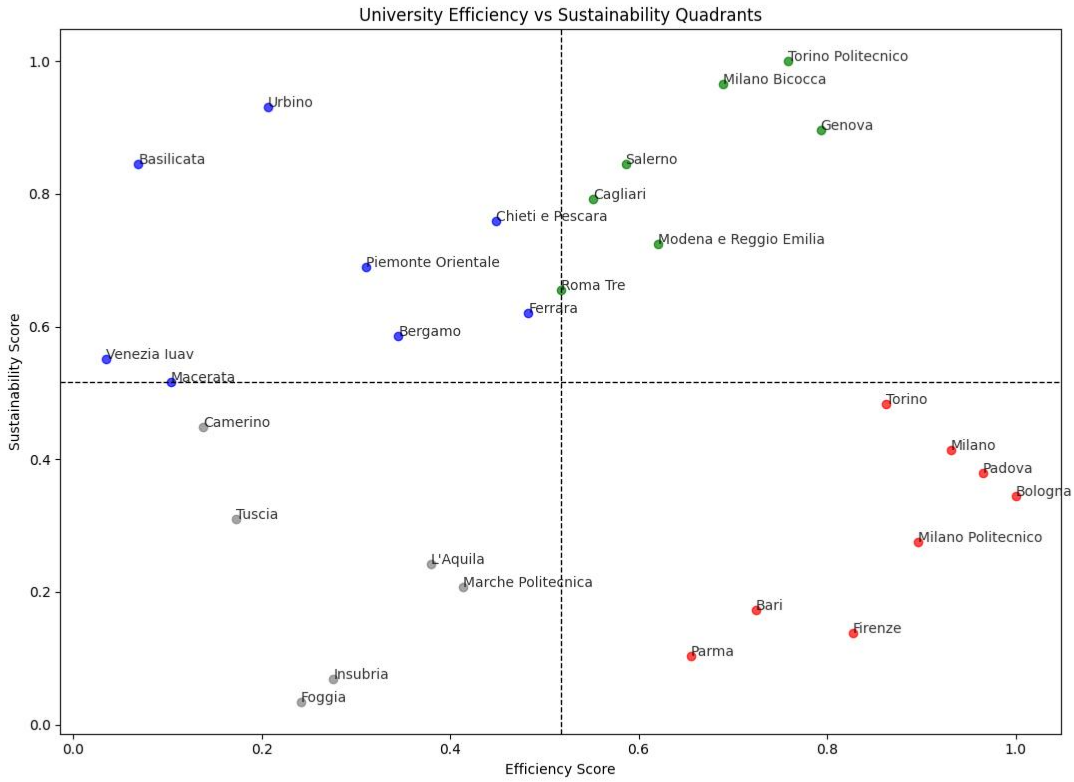

Figure 7: Institutional classification based on Technical Efficiency vs. Sustainability Score.

5.2 Differences in sustainability reporting behaviour

The qualitative comparison of sustainability reporting focused primarily on the thematic emphasis identified through word-frequency analysis and NLP processing. Rather than evaluating the formal quality or effectiveness of sustainability strategies, the analysis explored how different institutional groups emphasised sustainability-related themes within their accessible data.

Differences emerged across the four clusters. Universities in the High Efficiency–High Sustainability cluster frequently displayed stronger thematic emphasis on terms associated with governance, sustainability, research, and institutional measures. This suggests that sustainability language appears more consistently embedded within their institutional communication.

The High Efficiency–Low Sustainability cluster showed stronger concentration around themes linked to research, sustainability, training, development. It seems very close to the High Efficiency–High Sustainability cluster.

The Low Efficiency–High Sustainability cluster displayed more frequent emphasis on terms connected

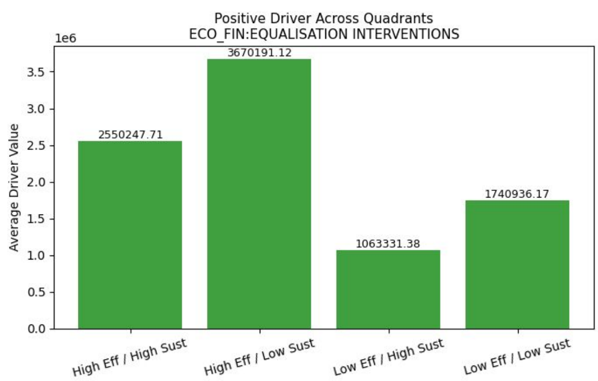

Figure 8: 'Equalisation Interventions' Average Value across Quadrants

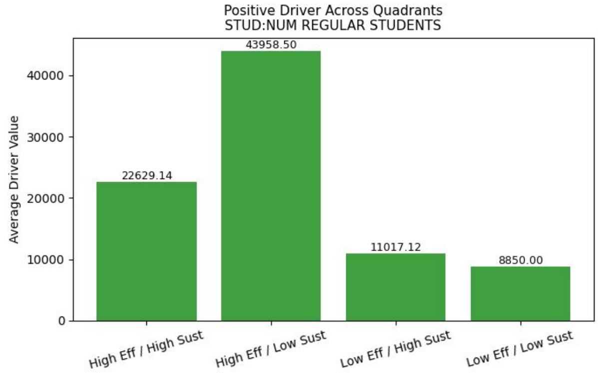

Figure 9: 'Num Regular Students' Average Value across Quadrants

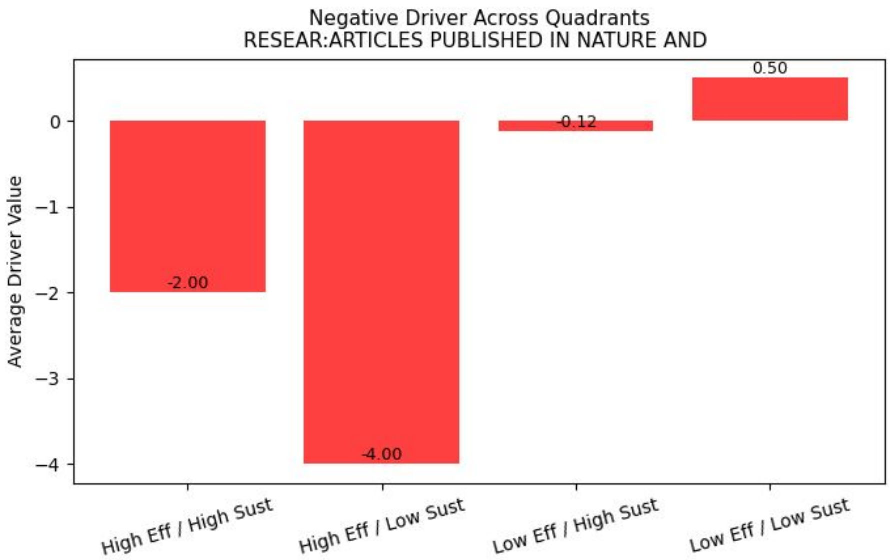

Figure 10: 'Articles Published in NAture and Science' Average Value across Quadrants

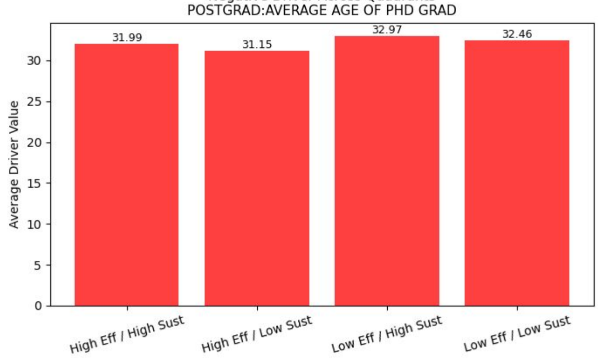

Figure 11: 'Average Age of PHD Grades' Average Value across Quadrants

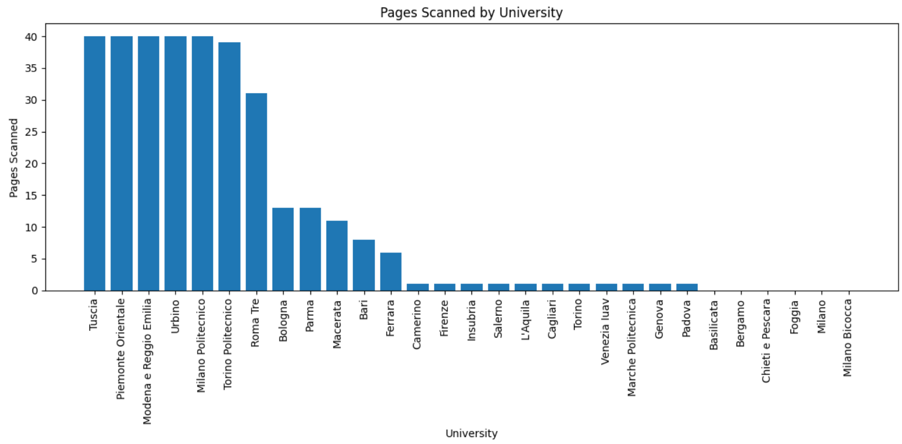

Figure 12: Pages Scanned by University

to inclusion, health, accessibility, and participation. This suggests that sustainability communication within these institutions is more strongly associated with social engagement and local impact rather than research competitiveness or operational scale.

Finally, institutions in the Low Efficiency–Low Sustainability cluster display unreliable patterns and focus less on recurring terms, suggesting terms such as skills, learning and knowledge.

The analysis reflects the visibility and thematic emphasis of sustainability-related language within publicly accessible institutional communication. Nevertheless, the observed differences suggest that universities frame and prioritise sustainability narratives differently depending on their broader institutional profiles and strategic orientation.

5.3 NLP as exploratory tools

The web scraping and NLP framework was used primarily as an exploratory tool designed to support institutional profiling rather than as an independent evaluative methodology. By collecting webpages, sustainability reports, governance documents, and SDG-related material, the framework enabled comparative analysis of reporting intensity and thematic emphasis across universities.

The NLP analysis identified recurring semantic differences between institutional groups. Research-intensive universities more frequently emphasised terms associated with innovation, competitiveness, and research output, whereas sustainability-oriented institutions showed stronger thematic concentration around inclusion, participation, territory, mobility, and social responsibility.

Importantly, the evidence intensity indicators generated through the scraping framework should not be interpreted as direct measures of sustainability performance. As illustrated by the distribution of scraped pages across institutions, reporting visibility varies substantially between universities. Some institutions, such as Tuscia, Urbino, Piemonte Orientale, and Modena e Reggio Emilia, produced a high number of discoverable sustainability-related pages, while others, including Basilicata, Bergamo, Foggia, Milano, and Milano Bicocca, returned little or no automatically discoverable content.

However, these differences do not necessarily indicate the actual absence of sustainability initiatives. In several cases, low evidence intensity may instead reflect fragmented website architectures, inaccessible PDF repositories, inconsistent URL structures. Also, universities with highly structured digital communication systems may appear disproportionately visible within the scraping framework despite not necessarily outperforming others in substantive sustainability outcomes.

For this reason, the evidence intensity indicators should be interpreted primarily as measures of institutional reporting visibility and digital discoverability rather than as direct indicators of sustainability quality or performance.

Overall, sustainability performance in Higher Education Institutions cannot be fully understood through quantitative rankings alone. Institutional narratives, governance structures, and reporting visibility all contribute to shaping how sustainability is operationalised and perceived within the higher education system.

6 Discussion

The findings of this study should be interpreted primarily as a proposed framework for bridging quantitative sustainability evaluation with qualitative institutional assessment, rather than as a definitive causal model of sustainability performance in Higher Education Institutions (HEIs). Building upon the methodological gap identified in the systematic review, the objective of this paper was not to develop the most mathematically sophisticated efficiency model, but instead to explore how relatively simple quantitative relationships can be transformed into interpretable institutional evidence capable of supporting qualitative analysis.

For this reason, the use of cross-correlation analysis was intentional. More complex non-linear approaches such as Data Envelopment Analysis (DEA) or stochastic frontier models may provide stronger predictive or optimisation capabilities, but they often reduce interpretability and widen the gap between quantitative outputs and institutional understanding. In contrast, the simplified approach adopted in this study allowed potential relationships between efficiency indicators and sustainability metrics to remain transparent and subsequently investigable through qualitative contextualisation.

The framework proposed in this paper therefore operates in two stages. First, quantitative relationships are used to identify potential institutional drivers and construct broad institutional groupings through the efficiency–sustainability quadrant system. Second, these outputs are explored qualitatively through exploring the behaviour of these drivers on the quadrants and analysing sustainability reporting visibility, thematic analysis, and institutional communication patterns obtained through web scraping and NLP techniques.

Importantly, the value of the identified drivers lies less in their statistical conclusiveness and more in their usefulness as exploratory institutional signals. Several of the positive drivers appeared relatively stable across multiple sustainability dimensions, while others, particularly among the negative drivers, appeared visually inconclusive when examined across quadrant distributions. This distinction itself reinforces one of the central arguments identified in the systematic review: purely quantitative sustainability evaluation risks overestimating weak or unstable relationships when institutional context is ignored.

Rather than treating the identified drivers as definitive causal mechanisms, this framework instead treats them as hypothesis-generating indicators capable of directing further institutional investigation. In this sense, the proposed methodology prioritises interpretability and qualitative investigability over mathematical optimisation.

6.1 Diagnostic usefulness of the quadrant framework

One of the central components of the proposed framework is the classification of universities into four institutional quadrants based on normalised efficiency and sustainability scores. Although initially developed as a visual comparative tool, the quadrant structure demonstrated broader diagnostic potential when interpreted qualitatively.

The framework suggests that universities should not necessarily be evaluated solely according to isolated ranking positions or individual quantitative outputs. Instead, institutions may benefit from understanding the broader organisational profile in which they are positioned.

The High Efficiency–High Sustainability quadrant represents institutions that appear capable of simultaneously maintaining operational productivity and strong sustainability alignment. Within the context of the proposed framework, these universities may function as useful reference cases for investigating how organisational coordination, sustainability visibility, and institutional strategy coexist within larger institutional structures.

The High Efficiency–Low Sustainability quadrant is particularly important diagnostically. These universities demonstrate relatively strong operational capacity and high performance across several positive efficiency-related indicators, yet comparatively weaker sustainability alignment. This suggests that sustainability limitations may not necessarily originate from resource scarcity or organisational inefficiency, but potentially from institutional prioritisation, governance focus, or sustainability integration processes. Consequently, this quadrant may represent institutions where sustainability improvements are operationally achievable without requiring major structural expansion.

The Low Efficiency–High Sustainability quadrant provides an important counterpoint to purely productivity-based interpretations of sustainability. In several cases, these institutions demonstrated comparatively stronger sustainability alignment despite lower overall efficiency performance. Furthermore, across many of the identified positive drivers, this group still appeared to perform more strongly than the Low Efficiency–Low Sustainability cluster. This may suggest that sustainability orientation is not entirely dependent on institutional scale or traditional productivity measures, and that organisational focus and institutional positioning may play a meaningful role in sustainability visibility and engagement.

Finally, the Low Efficiency–Low Sustainability quadrant appears to reflect institutions facing broader structural limitations across both operational and sustainability dimensions. However, these universities should not simply be interpreted as underperforming institutions. The framework instead suggests that they may represent organisations facing more substantial resource, infrastructural, or institutional constraints which cannot be adequately understood through rankings alone.

Overall, the quadrant system functions less as a ranking mechanism and more as a diagnostic framework capable of supporting institutional interpretation. Rather than identifying universally "good" or "bad" universities, the framework provides a structure through which universities may contextualise their institutional positioning and identify potential areas for strategic development.

6.2 Driver contextualisation through quadrant analysis

Following the identification of potential efficiency–sustainability drivers through cross-correlation analysis, the second stage of the proposed framework involved contextualising these relationships through the quadrant classification system. Rather than interpreting the identified drivers as isolated statistical outputs, the framework explored how selected drivers behaved across the four institutional groups defined by the efficiency–sustainability matrix.

To support interpretability, a subset of the strongest positive and negative drivers was visually analysed across the four quadrants using comparative bar plots. Importantly, the purpose of this stage was not to establish causal relationships, but rather to investigate whether the identified drivers displayed consistent or interpretable institutional patterns when viewed through the diagnostic grouping structure.

Across the positive drivers analysed, the High Efficiency–Low Sustainability quadrant consistently displayed the strongest values, followed by the High Efficiency–High Sustainability group. This pattern is particularly important because it suggests that strong operational productivity alone does not necessarily translate directly into sustainability alignment. In other words, universities with the strongest traditional efficiency-related indicators do not automatically appear within the highest sustainability grouping.

At the same time, for the majority of the positive drivers examined, the Low Efficiency–High Sustainability quadrant generally displayed stronger values than the Low Efficiency–Low Sustainability group. This may suggest that sustainability-oriented institutions still exhibit comparatively stronger organisational coordination or institutional focus despite operating at lower levels of overall efficiency. An exception to this pattern appeared within the equalisation-related measures, where the distribution across quadrants was less clearly differentiated. This may indicate that some institutional drivers operate differently depending on the organisational or financial structures underlying specific universities.

In contrast, the majority of the investigated negative drivers appeared visually inconclusive when compared across the quadrant distributions. Four of the five selected negative drivers did not demonstrate clearly interpretable separation between institutional groups. This finding is itself analytically relevant because it reinforces one of the central arguments of the proposed framework: purely quantitative identification of statistical relationships does not necessarily imply strong institutional interpretability.

However, one negative driver — publications in *Nature* and *Science* — displayed comparatively more distinguishable behaviour across the quadrant structure. Although exploratory, this pattern may suggest a potential tension between highly concentrated traditional research-performance incentives and broader sustainability alignment. Importantly, this should not be interpreted as evidence that research excellence negatively affects sustainability performance itself, but rather as an indication that institutions heavily oriented toward elite publication metrics may prioritise different organisational objectives within existing ranking structures.

The driver analysis therefore demonstrates how the proposed framework can be used not only to identify potential quantitative relationships, but also to qualitatively investigate whether those relationships appear institutionally meaningful when contextualised through diagnostic grouping structures.

Importantly, only a limited subset of the identified drivers was investigated directly within this study for interpretability purposes, while the remaining drivers are provided within the appendix. However, the framework itself is transferable and may be extended across additional drivers, alternative sustainability thresholds, or different institutional grouping strategies depending on the objectives of future investigations.

When evaluating these outputs, a critical distinction must be made between structurally credible drivers and statistical noise among the identified variables. For instance, the prevalence of **Postgradu-**

ate First-Level Master's Degrees functions as a highly credible positive driver, as curricular specialization directly reflects an expanded institutional capacity to deliver targeted sustainability education. Similarly, the negative relationship observed with **Articles Published in Nature and Science** serves as a credible strategic signal of an organizational trade-off, revealing that universities heavily incentivized to compete for elite bibliometric prestige may divert governance focus away from qualitative ESG reporting. In sharp contrast, variables like **Economic-Financial Equalisation Interventions** (*Interventi Perequativi*) and the **Average Age of Staff** clearly exemplify statistical noise amplified by high variable dimensionality and a small sample size ($N = 29$). Equalisation funding is fundamentally a redistributive safety net for structurally vulnerable institutions with no conceptual connection to environmental strategy, while staff age profiles merely reflect passive demographic patterns or historical hiring freezes rather than functional operational barriers. Ultimately, utilizing the quadrant matrix allows the framework to act as an analytical filter, separating these noise-driven coincidences from genuine signals of institutional priority.

6.3 NLP analysis and thematic institutional narratives

The NLP and word-frequency analysis operated primarily as an exploratory semantic profiling tool rather than as an evaluative methodology. Through word clouds, keyword frequencies, and SDG-related thematic detection, the framework enabled broad comparison of institutional communication emphasis across universities.

Importantly, these thematic patterns should not be interpreted as direct evidence of institutional sustainability performance. Instead, they provide insight into the language through which universities construct and communicate sustainability-related institutional narratives.

This component of the framework therefore complements the quantitative stage by introducing a qualitative layer of interpretability. While quantitative metrics identify broad relationships and institutional positioning, thematic analysis helps contextualise how sustainability is framed within institutional communication itself.

6.4 Bridging the quantitative–qualitative gap

The findings of this study reinforce the central argument identified in the systematic review: sustainability within HEIs cannot be fully understood through quantitative rankings or efficiency models alone. Quantitative evaluation frameworks inevitably simplify institutional complexity through the selection of measurable indicators, weighting systems, and ranking methodologies. While these models remain useful for comparative benchmarking, they may overlook the organisational, communicative, and governance dimensions through which sustainability is operationalised within universities.

The framework proposed in this study attempts to partially address this gap by treating quantitative outputs not as final evaluative conclusions, but as starting points for qualitative institutional investigation. Within this structure, drivers become qualitatively investigable signals, institutional quadrants become diagnostic categories, and reporting visibility becomes a contextual layer through which quantitative findings may be interpreted.

Ultimately, the study suggests that sustainability in higher education should not be understood solely as a measurable output, but rather as an institutional process shaped through the interaction between operational performance, communicative visibility, organisational priorities, and governance narratives.

6.5 Limitations

Several limitations must be acknowledged. First, the study relies primarily on exploratory correlation analysis, meaning that the identified relationships should not be interpreted causally. The relatively small sample size (29 HEIs) and the large number of institutional variables further limit the statistical generalisability of the findings.

Second, both the THE Impact Rankings and UI GreenMetric rankings are externally constructed evaluative systems that rely on their own methodological assumptions, weighting structures, and disclosure requirements. Consequently, sustainability scores themselves are not objective representations of institutional sustainability, but rather outputs of specific ranking methodologies.

Third, the web scraping framework measures reporting visibility and digital discoverability rather than operational sustainability itself. Variations in website structure, PDF accessibility, language formatting, and institutional communication practices may substantially influence the quantity of discoverable material.

Fourth, the NLP analysis relies primarily on keyword frequency and thematic matching rather than deeper semantic interpretation. As a result, the thematic findings should be interpreted as broad exploratory indicators rather than definitive institutional classifications.

Finally, the study is restricted to Italian universities and therefore reflects the institutional, regulatory, and reporting context specific to the Italian higher education system.

6.6 Future research

Future research could extend this framework in several directions. Larger multi-country datasets could improve the robustness and transferability of the proposed institutional classifications and driver analysis. Longitudinal studies may also allow investigation into how universities transition between efficiency–sustainability quadrants over time.

Additionally, more advanced quantitative approaches, including non-linear modelling, causal inference techniques, or machine learning methods, could be incorporated while preserving the interpretability-focused structure proposed in this paper. Future work may also benefit from richer qualitative methodologies, including interviews, governance analysis, or institutional case studies capable of validating the organisational mechanisms underlying the identified patterns.

Finally, more advanced NLP approaches, including semantic embeddings or transformer-based language models, could provide deeper analysis of sustainability narratives and institutional communication structures beyond simple keyword frequency analysis.

7 Conclusion

This study proposed an exploratory framework for bridging the divide between quantitative university rankings and qualitative institutional sustainability assessment within Higher Education Institutions (HEIs). Building upon the methodological gap identified in the accompanying systematic review, the objective was not to construct the most mathematically sophisticated efficiency model, but rather to demonstrate how quantitative sustainability relationships can be transformed into interpretable institutional evidence capable of supporting qualitative contextualisation.

Using Italian universities as a case study, the framework integrated institutional efficiency indicators from the Sapientia Observatory with sustainability metrics derived from the THE Impact Rankings and UI GreenMetric systems. Through exploratory cross-correlation analysis, the study identified a set of potential efficiency–sustainability drivers which were subsequently contextualised using a four-quadrant institutional classification framework and supported by qualitative analysis of sustainability reporting visibility, institutional communication, and NLP-based thematic exploration.

The findings suggest that strong operational efficiency does not necessarily imply strong sustainability alignment. In particular, several institutions displaying high levels of traditional academic productivity were positioned within the High Efficiency–Low Sustainability quadrant, indicating that sustainability performance may depend as much on institutional prioritisation and governance orientation as on resource availability or scale. Conversely, a number of lower-efficiency institutions demonstrated comparatively strong sustainability alignment and reporting visibility, reinforcing the idea that sustainability integration cannot be fully understood through productivity indicators alone.

The quadrant framework further demonstrated potential diagnostic usefulness by enabling institutional groupings to be interpreted qualitatively rather than simply ranked numerically. Rather than treating sustainability as a singular quantitative outcome, the framework highlights how operational efficiency, sustainability reporting, institutional narratives, and organisational priorities interact to shape broader sustainability positioning within HEIs.

Importantly, the qualitative stage of the framework reinforced one of the central conclusions of the systematic review: quantitative rankings alone risk oversimplifying institutional sustainability performance. The NLP and web-scraping analysis showed that sustainability reporting visibility and institutional communication vary substantially across universities, while also illustrating that digitally visible sustainability narratives do not necessarily correspond directly to operational sustainability outcomes. Sustainability communication therefore appears to function not only as a transparency mechanism, but also as a form of institutional positioning capable of shaping external perception within sustainability ranking ecosystems.

Overall, this paper contributes to the literature by proposing a structured framework through which quantitative sustainability evaluation can be linked to qualitative institutional interpretation. Rather than replacing existing ranking methodologies, the framework is intended as an interpretability-focused extension capable of supporting exploratory institutional diagnosis, contextual analysis, and future mixed-method sustainability research.

While exploratory and limited in scope, the study demonstrates that bridging quantitative and qualitative approaches may provide a more institutionally meaningful understanding of sustainability performance in higher education than either approach can achieve independently.

References

Bautista-Puig, N., Orduña-Malea, E., et al. (2022). Enhancing sustainable development goals or promoting universities? *International Journal of Sustainability in Higher Education*, 23(8):211–231.

Ezza, A., Pischedda, G., and Jongbloed, B. (2024). The disclosure of SDG performance: A comparative analysis of the dutch and italian higher education systems. In *Sustainability in Higher Education*, pages 183–202. Springer Nature Switzerland.

Fiorani, G. and Di Gerio, C. (2022). Reporting university performance through the sustainable development goals of the 2030 agenda: Lessons learned from italian case study. *Sustainability*, 14(15):9006.

King, E., Caçador, F., de la Rubia, N., and De Cicco, N. (2026). University efficiency, performance, and sustainability: A systematic review. Unpublished manuscript, Sapientia Observatory Project.

Lauder, A., Sari, R. F., et al. (2015). Critical review of a global campus sustainability ranking: Greenmetric. *Journal of Cleaner Production*, 108:852–863.

Lozano, R., Ceulemans, K., et al. (2015). A review of commitment and implementation of sustainable development in higher education: results from a worldwide survey. *Journal of Cleaner Production*, 108:1–18.

Malešević Perović, L. and Mihaljević Kosor, M. (2020). The efficiency of universities in achieving sustainable development goals. *Amfiteatru Economic*, 22(54):516–532.

Sapienza Università di Roma (2026). Sapientia observatory: Italian university efficiency and sustainability dataset. `https://www.uniroma1.it/it/pagina/portale-statistico-sapienza`. Internal Research Database, Sapienza University of Rome.

Times Higher Education (2025). The impact rankings. `https://www.timeshighereducation.com/`.

United Nations (2024). The 17 goals. Accessed: April 24, 2026.

Universitas Indonesia (2025). UI GreenMetric World University Rankings 2025. `https://uigreenmetric.com/rankings/university/overall-rankings-2025`. Accessed: 2026-05-12.